Supplementary Materials

Transfer learning from computed stability data for nanobody melting-temperature prediction

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Supplementary methods

Details of the heterogeneous MD data set

The heterogeneous data set was assembled from a SAbDab summary and the corresponding structures [1]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. This rule yielded 1143 samples with a usable sequence, topology, trajectory, and native-contact label, representing 833 unique amino-acid sequences. Ninety-seven sequences occurred in more than one PDB entry and accounted for 407 samples. We calculated native-contact values separately for each structure and sampled them individually during training; duplicate sequences were not collapsed, and the model did not receive the PDB identifier. Eleven sequences exceeded the 158-residue model input and were truncated at the end.

Structures were prepared and protonated at pH 7.4 with PDBFixer, pdb2pqr, and PROPKA [2, 3]. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [4, 5]. Equilibration at 300 K used 1 ns NVT and 1 ns NPT with a 2-fs timestep, after which two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps^{-1} friction [6].

The contact calculation used all backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the first production frame, as for the single mutation scans. Q was averaged over the final 30 ns of production, that is the last 300 frames of 100-ps output. The two MD data sets thus shared one Q definition, contact set, native reference, and averaging window, and differed only in the variants they covered.

FEP charge-change check

For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested the leading analytical periodic net-charge (Ewald self-energy) correction [7], $0.086(\Delta q)^2$ kcal mol^{-1}. We applied it and repeated the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels. Across the 284 charge-changing mutations the scaled labels barely moved, with a maximum shift of 0.008 on the $[0, 1]$ scale and a correlation of 0.99989 with the uncorrected labels, so the training labels use the uncorrected $\Delta\Delta G$.

Side-chain contacts in the native-contact label

The native-contact fraction Q in the main analysis uses backbone heavy atoms only. We also evaluated the side-chain-inclusive definition, in which all protein heavy atoms are eligible for contacts. On the matched 400 K mutation scan this raised the label spread by roughly 1.4-fold, but it did not improve transfer. With hyperparameters held fixed it slightly increased the held-out T_m MAE, and re-selecting the learning rate, dropout, and encoder learning rate on the 114-example validation set did not recover the backbone result. In the fine-tuned regime the side-chain label was significantly worse than the T_m-only baseline (paired ΔMAE +0.30 °C, 95% CI [+0.12, +0.49]), while the backbone label was neutral; in the frozen regime it erased the backbone label's benefit (ΔMAE from −0.20 to −0.04 °C). Because the mutant side chains were built without repacking their neighbors, side-chain contacts near each mutation carry construction noise rather than added signal, so the main analysis retains the backbone-only definition.

Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the Tm-only model used the shared architecture, dropout 0.05, and an encoder learning rate of 1×10^{-4}. The FEP model used the shared architecture with separate 1MEL and 4IDL heads, $n = 320$ samples per structure, dropout 0.15, and an encoder learning rate of 3×10^{-5}. Both conditions used a non-encoder learning rate of 3×10^{-4}, weight decay 0.04, and ten training seeds.

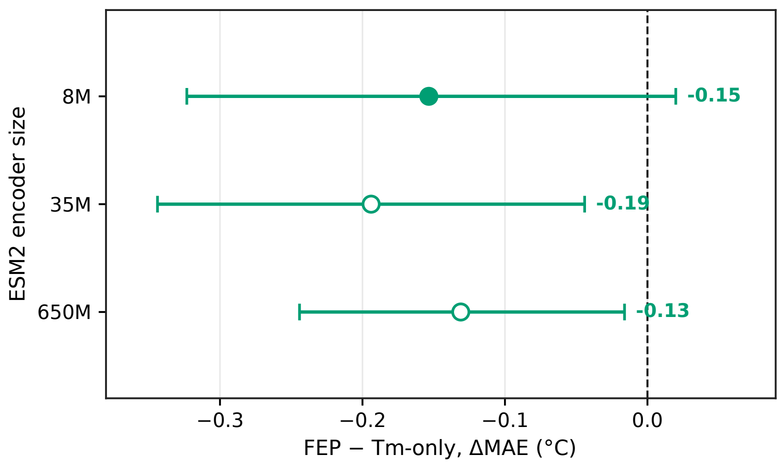

Figure S1 The FEP result held across ESM2 encoder sizes. FEP-minus-Tm-only MAE (ΔMAE) with a fine-tuned encoder; negative values mean lower Tm error with FEP. The 8M point is the five-seed model selected by the two-stage search, and the 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences.

References

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